

Battery Management System.

BMS

BY



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Linkedin post:-

[Mohmed Yousef on LinkedIn: After 40 days of hard working as a team with ...](#)

About BMS:- A Battery Management System (BMS) is an essential component in modern battery-powered devices, including electric vehicles, energy storage systems, and portable electronics. Specifically, in a 350V system, which is typical for many electric vehicles or large energy storage applications, the BMS plays a crucial role in managing the battery's performance, safety, and longevity. Here's an overview of how a BMS functions in such a system:

1. Voltage and Temperature Monitoring:

Voltage Monitoring: The BMS constantly monitors the voltage of each cell in the battery pack to ensure that they remain within safe limits. This is critical in a 350V system, where individual cells may be connected in series and parallel configurations.

Temperature Monitoring: The BMS tracks the temperature of the battery pack to prevent overheating, which can lead to thermal runaway, especially in high-voltage systems like 350V. Temperature sensors are placed strategically within the battery pack.

2. Balancing:

Cell Balancing: In a 350V system, cells are often connected in series, and their voltage may differ slightly due to manufacturing variations or different rates of aging. The BMS ensures that all cells are balanced, meaning their voltages are equalized, to maximize the overall performance and longevity of the battery pack.

3. State of Charge (SOC) Estimation:

The BMS estimates the State of Charge (SOC) of the battery, which represents the remaining capacity of the battery as a percentage. Accurate SOC estimation is vital for the efficient operation of electric vehicles or energy storage systems.

4. State of Health (SOH) Monitoring:

The BMS tracks the State of Health (SOH) of the battery, indicating the overall condition and remaining lifespan of the battery. This helps in predicting when the battery may need to be serviced or replaced.

5. Fault Detection and Protection:

Overcharge/Over discharge Protection: The BMS protects the battery pack from overcharging or over-discharging, which can damage the cells and reduce the battery's lifespan.

Short Circuit Protection: In the event of a short circuit, the BMS quickly disconnects the battery pack to prevent damage or safety hazards.

Thermal Protection: If the battery temperature exceeds safe limits, the BMS can reduce the load or shut down the system to prevent overheating.

6. Communication with Other Systems:

The BMS communicates with other components of the system, such as the vehicle's control unit or energy management system, to provide data on the battery's status and to receive commands. This ensures coordinated operation and efficient energy management.

7. Data Logging and Diagnostics:

The BMS logs data on the battery's performance, usage patterns, and any faults that occur. This data can be used for diagnostics, predictive maintenance, and improving the design of future systems.

8. Safety Features:

In a 350V system, the BMS incorporates various safety features to prevent dangerous situations, such as isolating the battery in the event of a fault or activating cooling systems if the battery overheats.

Applications:

Electric Vehicles (EVs): In EVs, a 350V BMS ensures the battery operates efficiently and safely, maximizing driving range and battery life.

Energy Storage Systems: For grid storage or renewable energy integration, a BMS in a 350V system ensures reliable and long-term operation.

The BMS is a critical component in ensuring the safe, efficient, and reliable operation of high-voltage battery systems, particularly those in the 350V range.

The Model:-

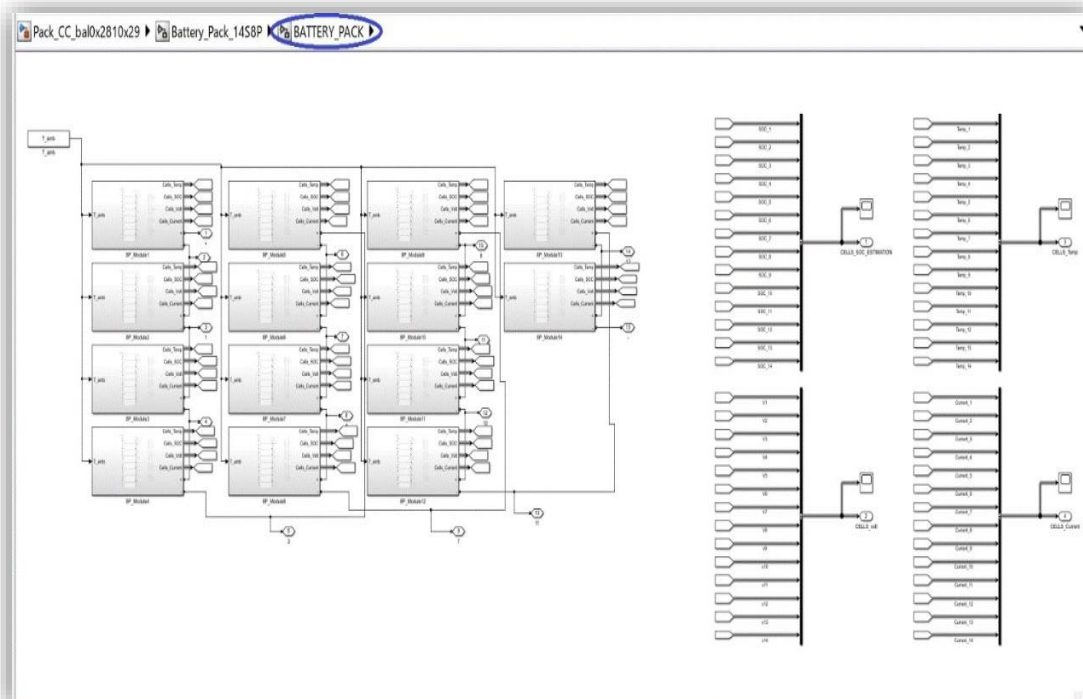
The battery pack.

we implement at first the battery pack mathematically to be 14S8P with content 112 cells with 20 Ah and average 1kw .

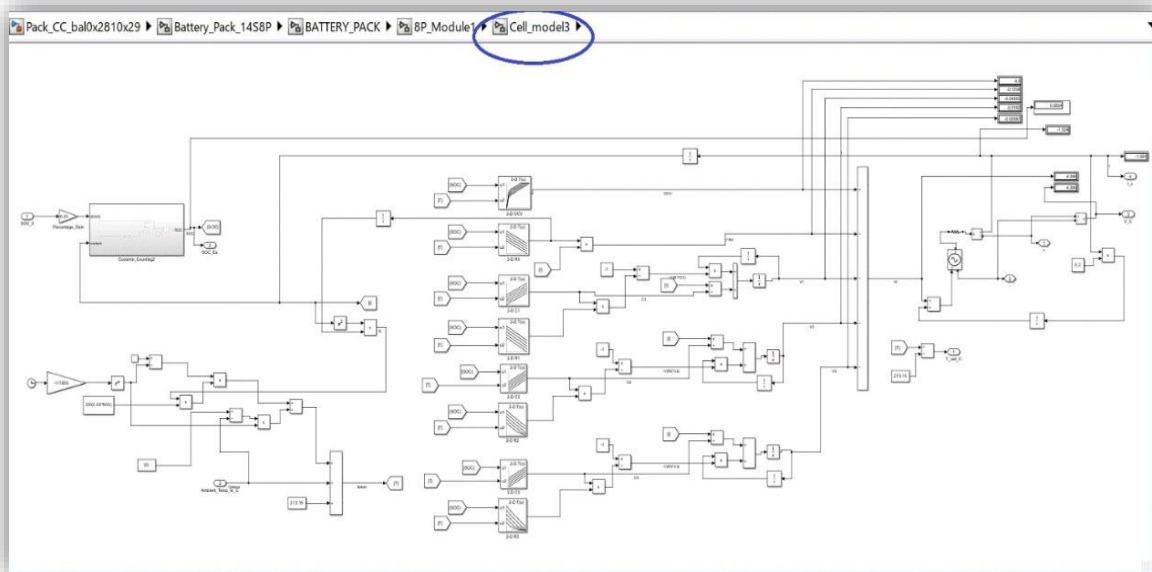
the cell model follow (3RC) pairs to represent the dynamics of a battery. This model is commonly used to simulate the electrical behavior of lithium-ion batteries and represents the temperature of a battery typically includes thermal dynamics to account for heat generation and dissipation within the cell. This model helps in predicting the temperature rise during operation, which is crucial for battery safety, performance, and longevity.

the pack consists of 112 cell every cell has a nominal voltage 3.7v and 2.6Ah.

the all pack:-



One cell:-



The feature of our model:-

- Soc estimation.
- balances cell voltages.
- monitor and protection.
- Calculate the available energy in the pack.
- Control the charge and the discharge and keep it running safely.

***Soc estimation.**

State of Charge (SOC) estimation is a key function of a Battery Management System (BMS). It represents the remaining capacity of a battery, usually expressed as a percentage, and is critical for predicting how long a battery can continue to operate before it needs recharging. Accurate SOC estimation is essential for the efficient and safe operation of battery-powered devices, especially in applications like electric vehicles and energy storage systems.

Methods for SOC Estimation

Several methods are commonly used to estimate SOC in a BMS, each with its own advantages and limitations:

***Coulomb Counting (Ah Counting):**

measuring the current flowing into and out of the battery over time. By integrating the current over time, the BMS calculates how much charge has been added to or removed from the battery.

Advantages: Simple to implement, real-time updates.

Limitations: Accumulates errors over time due to factors like sensor inaccuracies and battery aging, requiring periodic recalibration.

***Open Circuit Voltage (OCV) Method:**

The SOC is estimated based on the battery's open circuit voltage, which is the voltage measured when the battery is at rest (no load). OCV is correlated with SOC through a pre-established relationship specific to the battery chemistry.

Advantages: Accurate when the battery is at rest and fully relaxed.

Limitations: Not useful when the battery is under load, and the SOC-OCV relationship is non-linear and depends on temperature and battery age.

***Kalman Filter:**

A Kalman Filter is an algorithm that estimates the SOC by combining a predictive model of the battery with real-time measurements (e.g., voltage, current). It updates its estimates as new data comes in, effectively filtering out noise and improving accuracy over time.

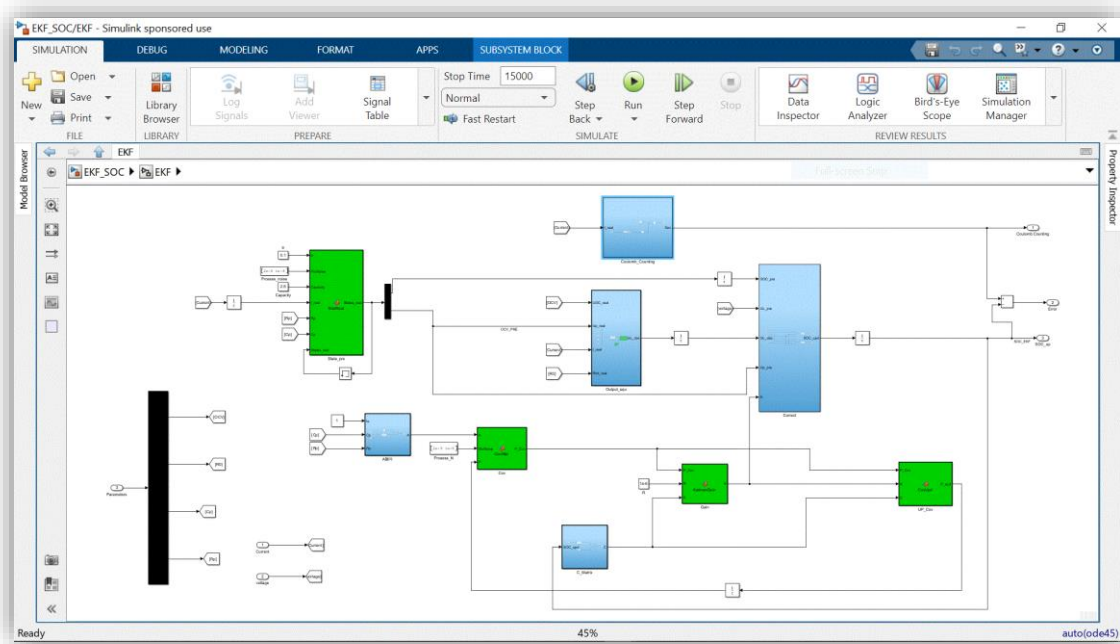
Advantages: High accuracy, adaptive to changes in battery conditions, and effective in real-time applications.

Limitations: Computationally intensive and requires a well-defined battery model.

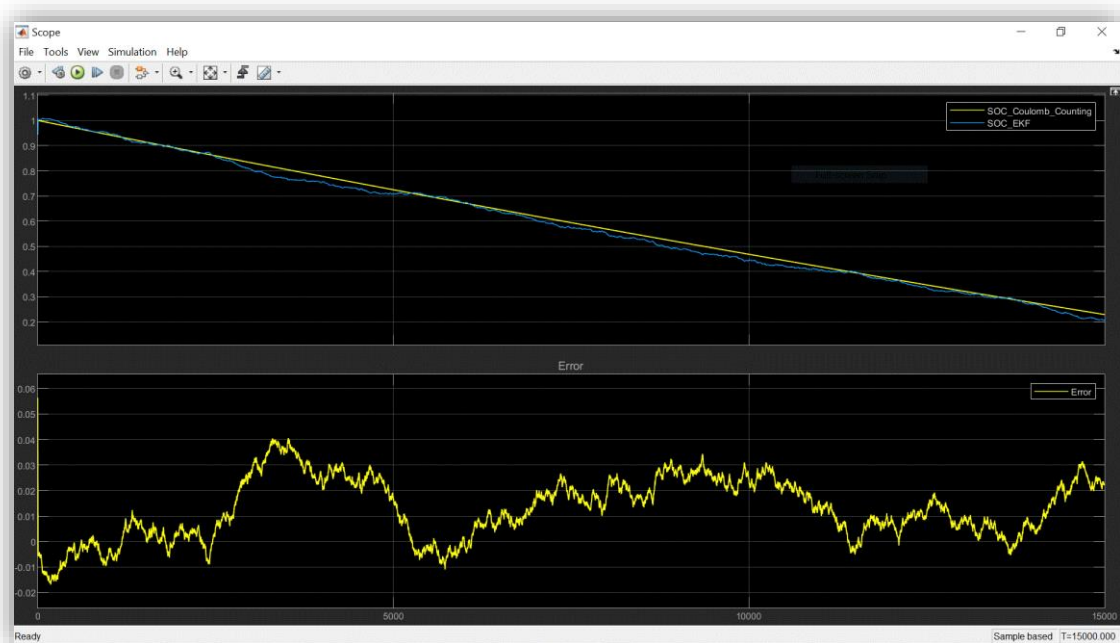
***EKF in the model.**

About it: <https://lnkd.in/dNtcjw5s>

Implementation:-



Result:-



***balances cell voltages.**

Cell balancing in a Battery Management System (BMS) ensures that all cells in a battery pack maintain the same voltage, which is crucial for maximizing performance, safety, and longevity. Cells in a battery pack can become imbalanced due to differences in manufacturing, aging, or operating conditions, leading to some cells becoming overcharged or over-discharged.

Types of Cell Balancing:

Passive Balancing:

Excess energy from higher-voltage cells is dissipated as heat through resistors until all cells reach the same voltage level. This method is simple and cost-effective but wastes energy.

Active Balancing:

Energy is redistributed from higher-voltage cells to lower-voltage ones using inductors, capacitors, or transformers. This method is more efficient as it conserves energy, though it is more complex and expensive.

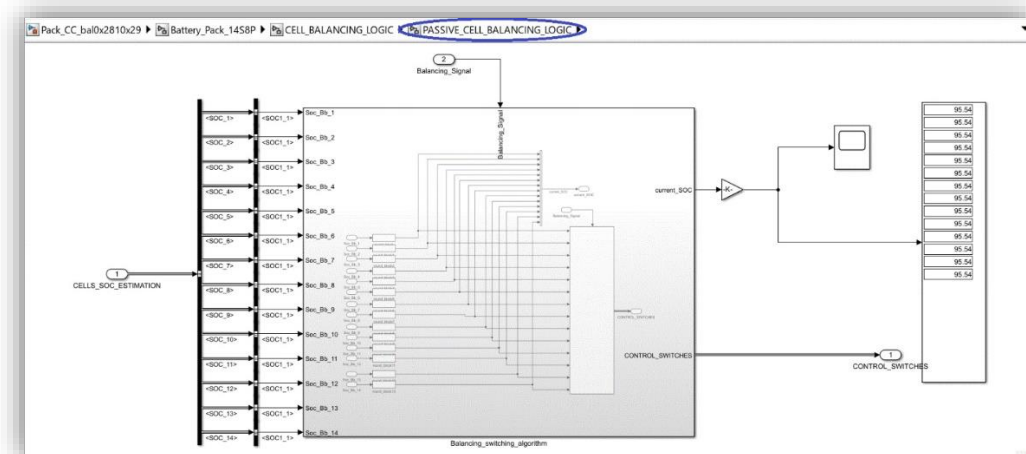
Benefits:

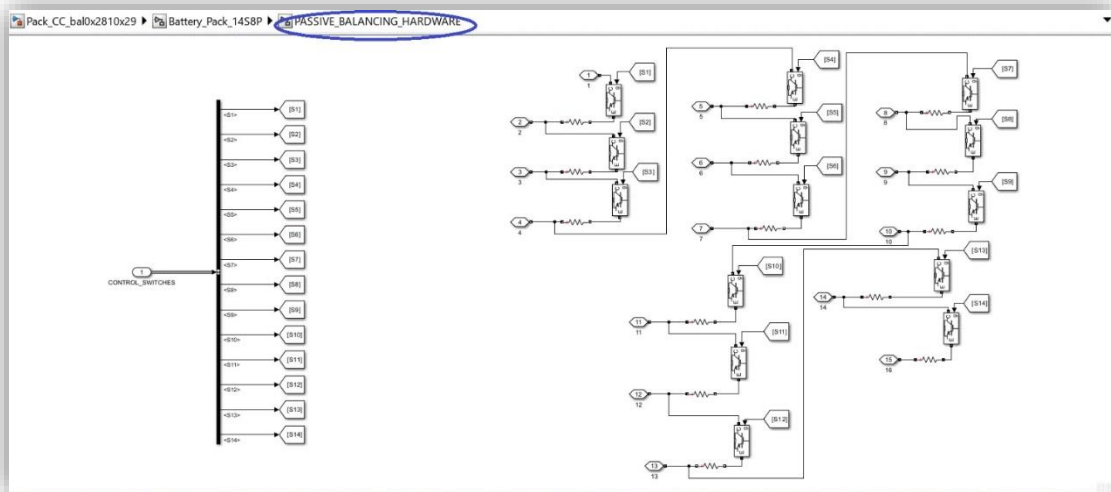
Enhanced Battery Life: Prevents individual cells from degrading prematurely.

Safety: Reduces the risk of overcharging or over-discharging, which can cause thermal runaway.

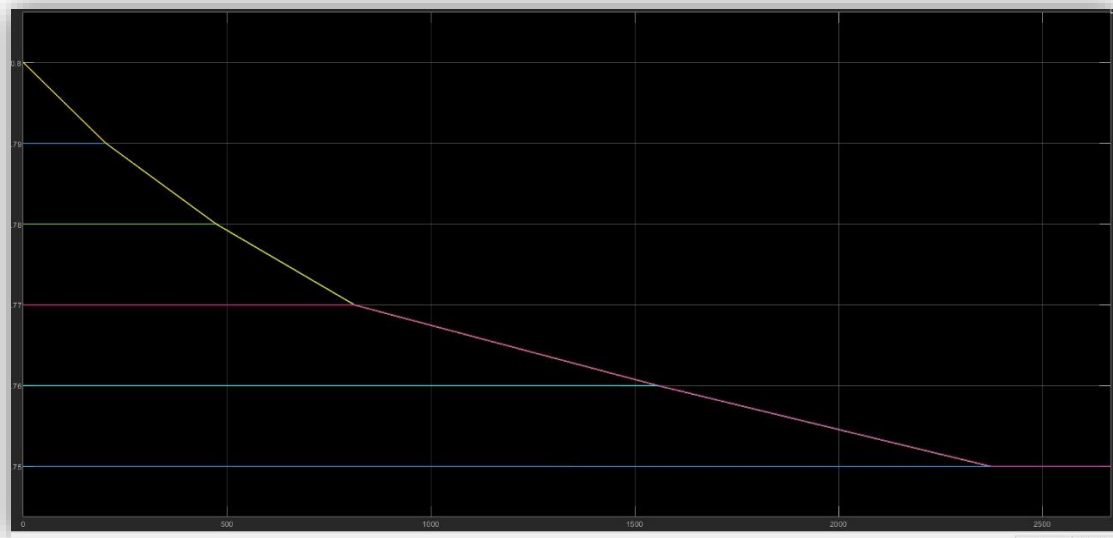
Performance: Ensures the battery delivers optimal power and capacity.

Implementation(Passive Balancing):-





Result:-



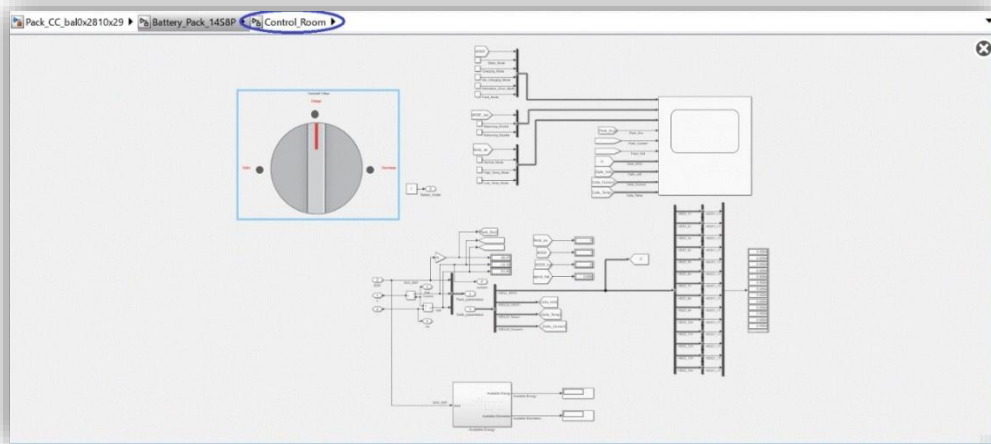
***Charging and discharging control.**

The pack can be in 3 modes charging ,discharging and static .

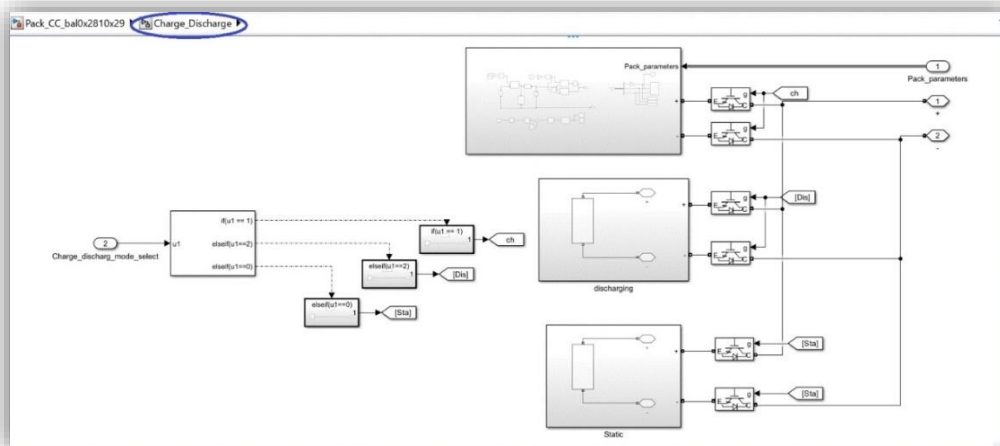
We have to control the modes to sure its go smoothly and safely so we make a something and called it **the control room** to handle it

Implementation:-

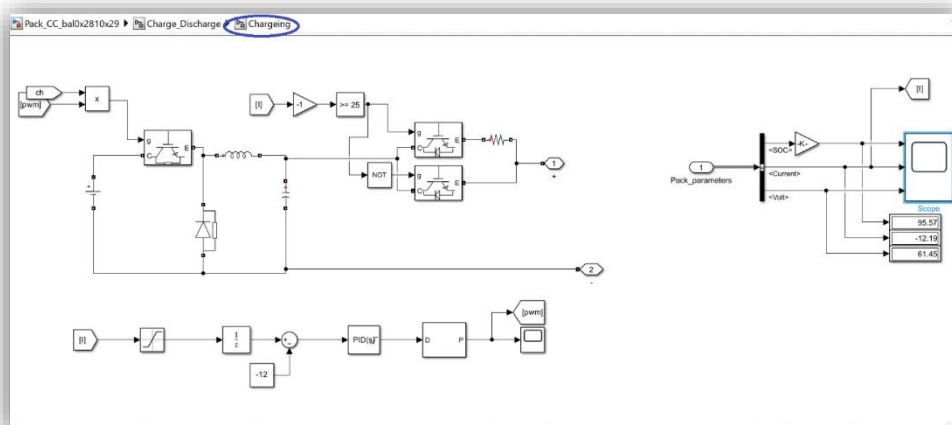
The control room:-



*Switch modes and contacts .

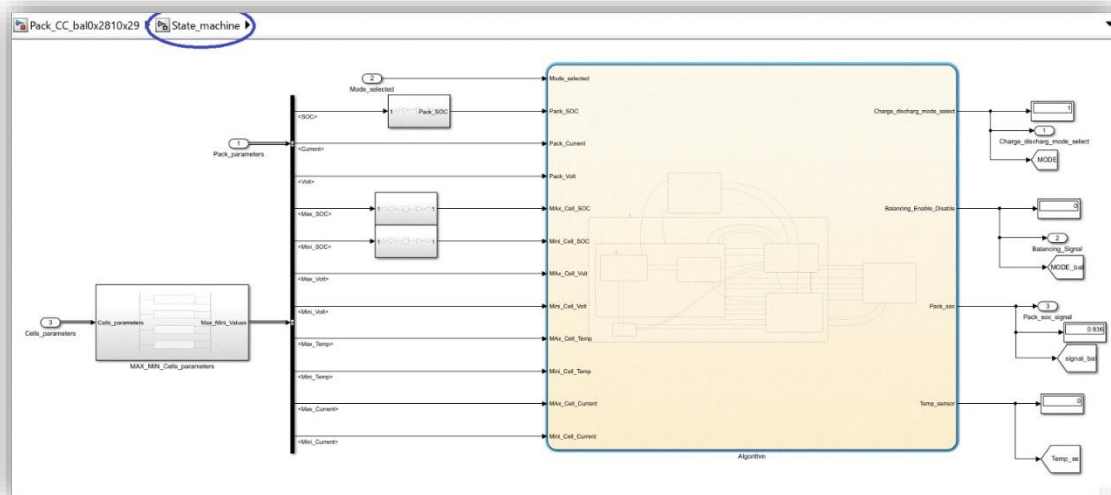


*Using a PID controller to keep the charging current within the suitable rated.



*The state machine(algorithm).

The state machine has an access for all the feature and it control the activation of it and monitoring the temperature and current in the safe zone and calculate the available power in the pack.



*The root level of the project.

